

# EXP028: Parameterized Non-Separable Quantum Walks on Tori

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## Abstract

This experiment extends the fixed non-separable coin results of EXP027 (Grover, DFT, Hadamard 4D) to a parameterized  $SU(4)$  coin framework on  $2 \times 2$  and  $4 \times 4$  tori.

On the  $2 \times 2$  torus, revivals occur only at  $\phi_R = 0$  and  $2\pi$ , shifting the fundamental period from 4 to 8 at  $\phi_R = \pi/2$ .

On the  $4 \times 4$  torus,  $\phi_R = \pi/2$  shifts the fundamental period from 12 to 24. Two-phase cancellation ( $\phi_R = \phi_U = \pi/2$ ) restores the original period of 12.

Biased coins (Right/Left only) revive on both tori at multiples of 4.

## 1 Introduction

In EXP027, we discovered full state revivals for non-separable quantum walks on tori using three fixed coins:

Table 1: EXP027 Fixed Coin Results Summary

| Coin        | $2 \times 2$ Torus       | $4 \times 4$ Torus |
|-------------|--------------------------|--------------------|
| Grover      | 4, 8, 12, 16, 20, 24, 28 | 12, 24             |
| DFT         | 16                       | None               |
| 4D Hadamard | 8, 16, 24                | None               |

## 2 Grover Coin

The Grover coin is:

$$G = \frac{1}{2} \begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix}$$

### 3 Results: $2 \times 2$ Torus

#### 3.1 Base Grover Coin (No Phases)

Fundamental period:  $N = 4$

#### 3.2 Single Phase $\phi_R = \pi/2$

Fundamental period shifts to  $N = 8$

Table 2: Phase sensitivity on  $2 \times 2$  torus

| $\phi_R$            | Period $N$ | Status     |
|---------------------|------------|------------|
| 0                   | 4          | Revival    |
| $0.11\pi - 1.89\pi$ | —          | No revival |
| $2\pi$              | 4          | Revival    |

#### 3.3 Two Phases: $\phi_R = \pi/2, \phi_U = \pi/2$

Cancellation restores period  $N = 4$

### 4 Results: $4 \times 4$ Torus

#### 4.1 Base Grover Coin (No Phases)

Fundamental period:  $N = 12$

#### 4.2 Single Phase $\phi_R = \pi/2$

Fundamental period shifts to  $N = 24$

Table 3: Phase sensitivity on  $4 \times 4$  torus

| $\phi_R$ | Period $N$ | Status     |
|----------|------------|------------|
| 0        | 12         | Revival    |
| $\pi/2$  | 24         | Revival    |
| $\pi$    | 12         | Revival    |
| $3\pi/2$ | 24         | Revival    |
| $2\pi$   | 12         | Revival    |
| Other    | —          | No revival |

#### 4.3 Two Phases on $4 \times 4$ : $\phi_R = \pi/2, \phi_U = \pi/2$

Cancellation restores period  $N = 12$

## 5 Comparison

Table 4: Comparison of phase effects

| Property                | $2 \times 2$ Torus        | $4 \times 4$ Torus                                 |
|-------------------------|---------------------------|----------------------------------------------------|
| Base Grover period      | 4                         | 12                                                 |
| $\phi_R = \pi/2$ period | 8                         | 24                                                 |
| Phase sensitivity       | Extreme (only 0, $2\pi$ ) | Moderate (0, $\pi/2$ , $\pi$ , $3\pi/2$ , $2\pi$ ) |
| Two-phase cancellation  | Yes                       | Yes                                                |

## 6 Conclusion

This experiment extends the fixed non-separable coin results of EXP027 to a parameterized framework on both  $2 \times 2$  and  $4 \times 4$  tori. The key findings, based on verified unitary coin implementations, are as follows:

### 6.1 Summary of Results

Table 5: Summary of parameterized coin revivals (fundamental periods)

| Torus        | Coin Configuration              | Period $N$ | vs Grover            |
|--------------|---------------------------------|------------|----------------------|
| $2 \times 2$ | Grover (fixed)                  | 4          | Baseline             |
| $2 \times 2$ | $\phi_R = \pi/2$                | 8          | Slower (2x)          |
| $2 \times 2$ | Biased (R,L only)               | 4          | Same                 |
| $4 \times 4$ | Grover (fixed)                  | 12         | Baseline             |
| $4 \times 4$ | $\phi_R = \pi/2$                | 24         | Slower (2x)          |
| $4 \times 4$ | Biased (R,L only)               | 8          | <b>Faster (1.5x)</b> |
| $4 \times 4$ | Biased (R,L) + $\phi_R = \pi/2$ | 4          | <b>Faster (3x)</b>   |

### 6.2 What the Parameterized Coin Achieves

1. **On  $2 \times 2$  torus:** No improvement over Grover. The fastest revival remains period 4.
2. **On  $4 \times 4$  torus:** A modest advantage exists. The biased Right/Left only coin revives at period 8, which is faster than Grover’s period 12. Adding a  $\pi/2$  phase to this biased coin further accelerates revival to period 4 – a **3x speedup** compared to the standard Grover coin.
3. **Phase sensitivity:** The  $2 \times 2$  torus revives only at  $\phi_R = 0$  and  $2\pi$ . The  $4 \times 4$  torus revives at  $\phi_R = 0, \pi/2, \pi, 3\pi/2, 2\pi$ , with periods alternating between 12 and 24.
4. **Two-phase cancellation:** Adding phases to both Right and Up ( $\phi_R = \phi_U = \pi/2$ ) cancels the phase effect on both tori, restoring the original Grover period.

### 6.3 What the Parameterized Coin Does NOT Do

- **Does not create new revival periods** on any torus (only shifts existing fundamental periods)
- **Does not produce revivals on new torus sizes** ( $3 \times 3$ ,  $3 \times 4$ ,  $5 \times 5$ , etc. remain without revivals)
- **Does not produce 11 revival periods** on  $4 \times 4$  – that result was an artifact of non-unitary coin implementations and has been corrected

### 6.4 Honest Assessment

The parameterized non-separable coin framework offers a **small, targeted advantage**: on the  $4 \times 4$  torus, a biased Right/Left only coin with a  $\pi/2$  phase achieves revival every 4 steps – three times faster than the standard Grover coin (period 12).

However, this advantage does not extend to other torus sizes. The  $2 \times 2$  torus is already optimal with the fixed Grover coin, and larger tori ( $3 \times 3$ ,  $3 \times 4$ , etc.) show no revivals regardless of parameterization.

### 6.5 Final Remarks

The parameterized coin framework successfully generalizes EXP027’s fixed coins, allowing tunable revival periods on the  $4 \times 4$  torus. The fastest revival discovered (period 4) represents a meaningful but modest improvement over the baseline Grover coin. This provides a tunable framework for Objective 4, though the scope of improvement is limited to small tori.

### Code Availability

[https://github.com/ianfromsligo/EXP028\\_parameterised\\_coin\\_tori\\_non\\_separable](https://github.com/ianfromsligo/EXP028_parameterised_coin_tori_non_separable)